

CONTACT

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GitHub: DagaVedant
Monroe Township, NJ

TECHNICAL SKILLS

Programming
Python, JavaScript, TypeScript, HTML/CSS

Frameworks & Tools
React, FastAPI, Flask, Git/GitHub, REST APIs

AI & Data
Machine Learning, Data Analysis, Statistical Modeling, Data Visualization

Engineering
Arduino, Raspberry Pi, IoT Systems, Embedded Systems, Sensor Integration

Leadership
Curriculum Development, Mentorship, Public Speaking, Program Management

ATHLETICS

– JV Tennis (1st Singles)
– JV Soccer

ACTIVITIES

– 4-H Innovation Club (Secretary)
– TSA (National Qualifier)
– NASA TechRise (Participant)
– Model United Nations
– Math League
– FBLA
– Leadership for Life

COMMUNITY SERVICE

600+ Volunteer Hours
Focused on STEM education, youth mentoring, and serving my community across New Jersey. Submitting for Congressional Award soon.

MARTIAL ARTS

– 1st Degree Black Belt in Taekwondo
– 3rd Place Sparring, NJTKD 2023

Vedant Daga

Machine Learning • Healthcare AI • Computational Biology • DataNova Research Competition Team

SUMMARY

I'm a student interested in using machine learning for immunology and disease detection, particularly through systems where biological data meets computational modeling. I'm an AP Biology student, a programmer, and a STEM and IoT instructor for the past two years. I've built tons of experience around biology, software development, and education. I'm extremely interested in applying my knowledge in real-world healthcare challenges to use my skills in research, project-based work, and collaboration.

EDUCATION

Edison Academy Magnet School | Edison, NJ

Expected Graduation: June 2029

Relevant Coursework: Electronics & Computer Engineering, AP Biology, Honors Algebra II

EXPERIENCE

Robbinsville 4-H Innovation Club | Secretary & Mentor

2022 – Present, Robbinsville, NJ

- I lead and mentor a group of student teachers and help them plan lessons, classes, and create material, teaching over 70 students and managing outreach events at schools, libraries, and senior centers.
- I co-designed and co-taught a 6-month IoT curriculum for students in India.

4H Summer Learning Program | Co-founder & Teacher

2026 – Present

- I'm co-founding a community STEM summer program as part of the Innovation Club, creating curriculum and helping plan out the structure.
- Currently in progress but almost ready to begin this summer.

PROJECTS

PulseFlow AI | Hospital Digital Twin Platform — HackJPS 2026, Best AI/ML Project

- I built a real-time hospital simulation platform modeling patient flow and predicting bottlenecks using simulation and optimization techniques.
- I integrated Google OR-Tools for staff and resource allocation optimization.
- I developed a Python/SimPy simulation engine running at 60x real-time speed.
- I built a FastAPI + WebSocket backend with a Next.js frontend featuring live hospital visualization, crisis simulation mode, and AI-assisted decision support.
- I designed a unified healthcare operations system combining simulation, forecasting, and optimization.
- I received the Best AI/ML Project at HackJPS 2025.

VeggieBuddy | AI Diet Recommendation Platform — HackJPS 2025, Best AI/ML Project

- I developed a web application that recommends diet-friendly restaurants based on user preferences.
- I built it using a React + TypeScript frontend and Python/Flask backend.
- I integrated Google Maps Places API for real-time location-based recommendations.
- I received the Best AI/ML Project at HackJPS 2025.

Python Learning Platform | Developer — In Development

- I designed a beginner-friendly interactive platform for self-paced Python learning.
- I built it using HTML, CSS, and JavaScript.

4-H Curriculum Redesign | Lead — 2024 – Present

- I restructured the club's teaching framework to improve learning progression and engagement.
- I implemented a peer-teaching model where senior students mentor newer participants.

HONORS & AWARDS

- ★ Best AI/ML Project, HackJPS 2025
- ★ VEX IQ Worlds, 2025
- ★ 4th Place Individual & 3rd Place Team, MCAMC 2025
- ★ Top 10 Overall, MakeNJIT
- ★ TSA TEAMS Nationals
 - ★ 3rd Place TEAMS Multiple Choice Questions
- ★ Edison Academy Tech Expo Winner
- ★ Honorable Mention, ILMUNC 2026

PRESENTATIONS

- I presented an award-winning STEM project at the Edison Academy Advisory Dinner (selected by school administration).
- I represented Edison Academy at Middlesex County Day, presenting to the county-level.
- I ran STEM workshops across schools, libraries, and community centers through the Robbinsville 4-H Innovation Club.